MUHAMMAD HASAN FAROUK

Giza, Egypt (+20) 115 852 2627 Email
Portfolio LinkedIn GitHub

Professional Summary

Recent Computer Science graduate specializing in backend engineering, with a strong background in high-performance computing (HPC) and scientific programming. Experienced in building scalable RESTful APIs, managing relational databases, and developing robust architectures using **Java** and **Spring Boot**. Leveraging system-level optimization techniques, I focus on writing clean, maintainable code and engineering secure, highly efficient, and resource-conscious enterprise backend systems.

Education

Cairo University Giza, Egypt
B.Sc. in Computer Science Sep 2022 – July 2026
GPA: **3.36 / 4.0**

Technical Skills

- **Languages:** Java, Python, C++, SQL
- **Backend Frameworks:** Spring Boot, Spring Security, Hibernate / JPA, RESTful APIs, JWT
- **Databases & Cloud:** PostgreSQL, Redis, Microsoft Azure, Docker, Linux
- **Developer Tools:** Git / GitHub, Maven, Postman, Swagger / OpenAPI

Software Engineering Projects

Yaqazah – Cloud-Native AI Driver Monitoring Platform Feb 2026 – Jul 2026
Java 21, Spring Boot, PostgreSQL, Redis, Microsoft Azure, Git Giza, Egypt

- Architected a monolithic RESTful API utilizing **Java 21** and **Spring Boot**, documenting API contracts using **Swagger/OpenAPI**.
- Engineered secure and scalable data persistence with **PostgreSQL** and **Spring Data JPA**, leveraging **Redis** for session caching.
- Implemented robust stateless **JWT authentication** and customized Role-Based Access Control (RBAC) via Spring Security for isolated access.
- Deployed and maintained cloud infrastructure on **Microsoft Azure**, utilizing Git for version control and continuous integration workflows.

Learning Management System (LMS) Backend Jan 2025 – Jul 2025
Java 17, Spring Boot, PostgreSQL, Spring Security, Maven Giza, Egypt

- Designed a comprehensive LMS backend applying layered architecture (Controllers, Services, Repositories) for course management.
- Engineered **JWT** authentication and strict role-based access control (RBAC) for isolated user domains (Student, Instructor, Admin).
- Modeled **PostgreSQL** schemas, optimizing retrieval with **Spring Boot Cache** and reducing boilerplate code using **Lombok**.
- Integrated **Spring Boot Mail** for automated system notifications and structured standard CRUD operations ensuring high maintainability.

Scientific Programming & Experience

INFN-SESAME Efficient Scientific Computing International School Jan. 31 – Feb. 6, 2026
International Scholar (Fully Funded by INFN, SESAME & ASREN) Allan, Jordan

- Completed a rigorous training and examination program (25h lectures, 8h hands-on) led by **CERN and INFN** instructors.
- **Parallel & GPU Computing:** Optimized scalable applications using **MPI** (inter-node), **Intel TBB** (task-based), and **CUDA** kernels, maximizing occupancy and shared memory to accelerate computational workloads.
- **C++ Memory Engineering:** Applied modern C++ (RAII, smart pointers) and cache-aware programming (AoS/SoA) to mitigate false-sharing.
- **Performance Profiling:** Analyzed computational bottlenecks to optimize CPU/memory utilization and built Python bindings for C++ applications to integrate high-performance native code into scientific workflows.

Systems Engineering & HPC Intern July 2025 – Sept 2025
HPC R&D Laboratory, Faculty of Computers and AI, Cairo University Giza, Egypt

- Collaborated with an academic research team to successfully rebuild and configure the HiPer-FC computational cluster from isolated hardware components.
- Configured Linux environments and automated system deployments using SSH and shell scripting, ensuring secure access and a reliable computational infrastructure.